

TEZPUR UNIVERSITY
Design and Analysis of Algorithms (CO 206)

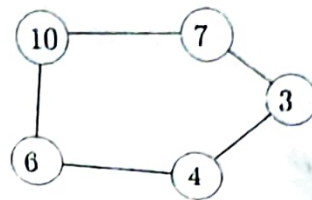
Date : May 22, 2022

Maximum Marks : 25

Duration : 60 minutes

You may answer all the questions but maximum you get is 25.

1. Consider the following problem. Given an n length cycle $C_n = (V, E)$ and a nonnegative weight $w(v)$ for each vertex $v \in V$, find the independent set $S \subseteq V$ of G with the maximum-possible sum $\sum_{v \in S} w(v)$ of vertex weights. In the figure below, the weight of the maximum independent set is 14.



- a) Deduce the total number of different independent sets in a cycle C_n . [8]
 - b) Derive the recursive structure of the optimal solution. [7]
 - c) Write the pseudocode for the dynamic programming solution of the above problem. The output of the algorithm should be the maximum weight and also the vertices of the independent set. [10]
2. If we modify Dijkstra's solution for single source shortest path problem in the following way.
- a) Instead of Min-heap store the score of the vertices in Max-heap.
 - b) Rather than calling Extract-min(), use Extract-max() operations.
 - c) update the score of a vertex iff the score increases.

the modified Dijkstra's algorithm will find the longest path of every vertex from source vertex. Prove or disprove. [10]